



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Assignment, Laboratory & Office Hours Manual

Evidence templates for the current Classroom and project workflow

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ACADEMIC YEAR	2026-2027 Academic Year
DOCUMENT ID	AU-OAA-ESET301-LAB-2026-02
REVISION	3.0 / v16.3.64
STATUS	Current Governing Document

DISCERE • AEDIFICARE • SERVIRE

Learn. Build. Serve. | Current runtime: v16.3.64 | Core curriculum release: 16.3

CURRENT WEBSITE USE - v16.3.64

Templates are optional documentation aids. Completion is recorded by the live Classroom / Lab / Assessment system. No separate hours log, spreadsheet gradebook, or manual mastery rating is required.

1. Evidence Standard

Every meaningful lab or project should leave enough evidence for you to reconstruct what you expected, what you did, what you measured, what failed, and why your final conclusion is credible.

- Prediction before measurement whenever the task allows it.
- Circuit/code/setup identity: schematic, wiring, pin map, configuration, or version reference.
- Measured or observed result with units, instrument settings, and relevant uncertainty/limitations.
- At least one troubleshooting note when something unexpected happens.
- A concise conclusion tied to the stated learning objective, not just "it worked."

2. Current Lab Completion Rule

LAB / APPLICATION

When a week has a lab, the shared completion truth requires a recorded physical or virtual route plus the required lab knowledge-check score of at least 80%. Physical proficiency is credited only when physical evidence exists.

3. Optional Learning Record

Field	Record
Week / module	
Date	
Topic	
What can I explain without notes?	
What did I calculate or code?	
What did I build or measure?	
What failed and what did I learn?	
Evidence location	
What should I review or repair next?	

4. Lab Evidence Sheet

Field	Record
Lab ID / title	
Route: physical or virtual	
Objective	
Pre-power / safety check	

Field	Record
Prediction / expected values	
Equipment / software / versions	
Procedure summary	
Measured / simulated results	
Percent difference / error where meaningful	
Fault / anomaly	
Root cause or best-supported hypothesis	
Fix / verification	
Evidence files / photos / captures	

5. Root-Cause Troubleshooting Record

Step	Prompt
1. Reproduce	What exact symptom can you reproduce?
2. Define expected	What should the node/signal/software state be?
3. Measure	What measurement or observation discriminates between likely causes?
4. Isolate	What boundary, block, net, interface, or state did you narrow it to?
5. Root cause	What evidence supports the cause? Distinguish cause from symptom.
6. Correct	What did you change?
7. Verify	What fresh test proves the fix and checks for regressions?
8. Document	What would another technician need to repeat the diagnosis?

6. Project 1 Design Review Template

- Problem statement and success criteria
- Requirements with measurable acceptance conditions
- System block diagram
- BOM and power budget
- MCU/peripheral selection and pin map
- Interface choices and expected signals
- Firmware architecture / state machine
- Fault and timeout strategy
- Bring-up sequence

- Test plan and PASS/FAIL criteria
- Known risks / scope cuts
- Evidence repository structure

7. Test Report Template

Section	Minimum content
Purpose	What question did the test answer?
Configuration	Hardware, firmware, software, revision, power, instruments.
Method	Repeatable steps and conditions.
Expected result	Numerical/behavioral prediction and tolerance where appropriate.
Observed result	Measurements, captures, logs, photos, or PASS/FAIL table.
Failures	At least the meaningful anomalies and their disposition.
Conclusion	What is supported, what remains uncertain, what action follows.
Attachments	Plots, serial logs, logic captures, oscilloscope screenshots, photos, code revision, issue links.

8. Role-Proof Dossier

By Week 29, collect concise, interview-defensible evidence across instrumentation, solder/rework/harness work, hardware bring-up, troubleshooting, and documentation. The dossier should show what you personally did and what tools you used.

9. Office Hours / Tutor Request Template

Bring	Example
Context	Week, lesson, lab, standard, or job posting.
Exact problem	Equation, circuit node, compiler error, waveform, logic capture, failed check, or confusing paragraph.
What you expected	State your current model first.
What happened	Give the actual result, including units/messages.
What you already tried	Avoid repeating dead ends.
What kind of help	Explanation, hint, practice question, debugging plan, or rubric review.

10. Current Lab Route Index

Week	Lab	Instructional context
W01	LAB-001	Electricity, Safety, Ohm's Law, and Power
W02	LAB-002	DC Networks, Kirchhoff's Laws, and Voltage Dividers
W03	LAB-003	DMM, Bench Supply, and Oscilloscope
W04	LAB-004	AC Waveforms, Capacitors, Inductors, and Time Response
W05	LAB-005	Foundation Mastery and Practical Fault Isolation
W06	LAB-006	Reactance, Impedance, Resonance, Filters, and Transformers
W07	LAB-007	Components, Magnetism, Relays, Motors, and Generators
W08	LAB-008	Diodes, BJTs, MOSFETs, and Controlled Switching
W09	LAB-009	Rectifiers, Filters, Regulators, and Power-Supply Faults
W10	LAB-010	Amplifiers, Op-Amps, Gain, Decibels, and Loading
W11	LAB-011	Number Systems, Boolean Logic, and Digital Circuits
W12	LAB-012	Soldering, ESD, Rework, and Version-Controlled Repair
W13	LAB-013	Schematics, Datasheets, Signal Flow, and Root Cause
W14	LAB-014	C Fundamentals and Computer Architecture
W15	LAB-015	Embedded C, Bit Operations, State Machines, and Timing
W16	LAB-016	STM32 Project Setup, GPIO, Clocks, and SWD Debug
W17	LAB-017	UART Communication and Python Serial Control
W18	LAB-018	I ² C, SPI, Logic Analysis, Sensors, and Actuators
W19	LAB-019	Cabling, Audio/Video Signal Paths, and Optical Systems
W20	LAB-020	RF, Telecommunications, Spectrum, and Technician Procedures
W21	LAB-021	CETa Integration I and Mixed Technician Decisions
W22	LAB-022	CETa Repair II and Project Requirements

Week	Lab	Instructional context
W27	LAB-023	Python Hardware Test, Logging, and CETa Computer-Application Retention
W30	LAB-024	Portfolio, Resume, Interview Defense, and CETa Retention

11. Safety and Integrity

- Do not treat a copied tutorial as independent evidence.
- Keep personal bench work within low-voltage/current-limited boundaries unless a qualified environment explicitly provides something else.
- For oscilloscope, mains, battery-energy, soldering/rework, and ESD hazards, follow authoritative procedures and equipment documentation.
- Preserve failed attempts when they explain the engineering process; do not manufacture clean results after the fact.